Donghyung Lee

DevOps Engineer

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Introduction

Hello, I am Donghyung Lee, a DevOps Engineer who designs infrastructure from the ground up and leads its transitions.

Rather than maintaining what was already there, I have mostly led greenfield builds and large-scale migrations. Starting with an OpenStack-based private cloud for financial and public-sector clients, I went on to lead the infrastructure side of a company-wide AWS → GCP migration at a startup, cutting cloud cost 50%. Today I am the sole owner of the AWS + on-premises hybrid infrastructure at a game studio, where I cut monthly infrastructure cost 20% and solely built and launched the EKS production environment for a new game.

My main strength is CI/CD automation, with observability layered on top for production stability. I have built CI/CD pipelines on GitLab CI, GitHub Actions and Jenkins with ArgoCD, and today on Kubernetes I fully automated build and deployment with GitLab CI and ArgoCD GitOps, cutting deployment time 67% (30min → 10min). New components are 100% codified with Terraform and Ansible (excluding IAM). On top of that, a 3-signal observability stack connects logs, metrics and distributed traces in Grafana, so an incident can be traced to its cause from a single view.

Beyond that, I build in-house DevOps tools myself to remove operational bottlenecks, and I have worked on AI/ML workload infrastructure as well. At the startup I standardized and scaled an image-generation GPU fleet across 10 in-house boxes, GCP A100 and L4, and 20 external-IDC servers; at the game studio I built an in-house LLM inference service on an on-prem GPU server, with a coding model kept loaded in GPU memory for the dev team.

I set myself the principle that systems and documents — not one person — should do the work. I believe infrastructure is finished not when it ships but when a fellow developer has a guide they can use on their own, and I keep building environments where solid automation lets the whole team move fast with confidence.

Strengths

Asking 'Why' First
- Ask why the work is needed, define the business problem, then pick the tool

Systematic Documentation & Knowledge Sharing
- Document and share every task, building organizational knowledge assets

Record-Based Continuous Improvement
- Record incidents, infra changes and troubleshooting as postmortems the team reviews together — preventing recurrence, driving improvement

Full-Stack Infrastructure Experience
- End-to-end infra across cloud (AWS, GCP), on-premises, networking, CI/CD, and monitoring

Skills

Cloud	AWS, GCP, OpenStack
Container Orchestration	Kubernetes, Docker, Cilium CNI, NGINX Gateway Fabric / Gateway API, Karpenter
Infrastructure as Code	Terraform, Ansible, Helm, Helmfile
CI/CD	GitHub Actions, GitLab CI/CD, ArgoCD, Argo Rollouts, Jenkins, GitOps
Monitoring & Observability	PLG Stack (Prometheus/Loki/Grafana), ELK Stack, EFK Stack, ECK Operator, Fluent-bit, Thanos, OpenTelemetry, Grafana Tempo
Security & IAM	Keycloak (Operator, OIDC IdP), Vaultwarden, GitLab SSO (OIDC), External Secrets Operator, Sealed Secrets
Automation	Go, Python, Shell Scripting
Database	MySQL, PostgreSQL, Redis, MongoDB
OS & Servers	Ubuntu, Rocky Linux, CentOS, Debian, GPU Server

Professional Experience

Phantom (Concrit Studio)

2024.10 ~ Present

DevOps Engineer

Sole DevOps infrastructure engineer at a game studio, designing, building, and operating AWS-based cloud and on-premises servers. Codified infrastructure with Terraform/Ansible (100% IaC for new components, excluding IAM) and documented operations via automation scripts and postmortems.

- Cut monthly infra cost 20% — sole design and build of the AWS + on-prem hybrid, 100% IaC on new components (excluding IAM)
- Sole-built the launch infrastructure for a new game, expanding a single on-prem dev cluster into an on-prem + AWS multi-cluster (EKS platform 100% declared in Git excluding IAM, Karpenter autoscaling). Moving onto an EKS Auto Mode cluster then cut the compute layer's Terraform-managed resources by 83% (102 → 17) and took over production, retiring the old cluster (DB moved in 53s, all 36 tables verified); a WireGuard site-to-site VPN gave on-prem a private route
- Built Metrics · Traces · Logs 3-signal observability — log search on one Kibana interface (a 3-stage redesign: ELK → EFK → ECK Operator), OpenTelemetry distributed tracing (OTel Collector · Tempo) and kube-prometheus-stack
- Extended the EFK log pipeline into a product-analytics platform — ES Transform aggregating events per user and Kibana serving D+1~D+30 retention and cohorts, with no commercial analytics SaaS
- Automated manual build-and-deploy with GitLab CI + ArgoCD pull-GitOps, cutting deployment time 67% (30min → 10min); shared toolchain images cut CI provisioning ~97% (5min → ~10s), plus Argo Rollouts Canary and infra automation ported to Python
- Consolidated multiple ingress-nginx instances onto NGINX Gateway Fabric, central SSO for Harbor and ArgoCD via Keycloak Operator, plaintext AWS production DB passwords removed via External Secrets Operator, static IAM keys removed via IRSA/Pod Identity and static SSH keys via SSM
- Built solo over 6 weeks a self-service portal placing 7 internal apps behind a single OIDC authentication gate, plus 5 operations services (accounts · Keycloak · Harbor · Elasticsearch · VPN) wired to 10 internal account systems, with permissions scoped per team — operational requests that had to pass through one engineer are now handled by each team directly, within its own permissions

Key Achievements

- 20% Monthly Infra Cost Reduction (savings reinvested in new projects)
- Greenfield EKS Sole-Built Production Launch Infra (single on-prem dev cluster → on-prem + AWS multi-cluster, serving real traffic since the 2026.07 soft launch)
- 67% Deployment Time Reduction (30min → 10min, faster market response)

NerdyStar

2023.03 ~ 2024.07

DevOps Engineer

Led the infrastructure side of a large-scale AWS → GCP migration at a game/blockchain startup and built GitOps CI/CD pipelines.

- AWS → GCP large-scale migration (Shared VPC + Workload Identity Federation keyless GitHub Actions auth, subdomain-delegated DNS cutover during a maintenance window, Cloud Armor + Cloud CDN, self-hosted NFS replacing Filestore)
- Built GitOps CI/CD pipeline on GitLab CI · GitHub Actions · Jenkins · ArgoCD
- Built game/blockchain data pipeline (MongoDB · CloudSQL · GA · Dune → BigQuery → Google Sheets, Cloud Functions + Scheduler + Dataflow, Terraform IaC) — replacing analysts' 2-3 hours of daily manual collection with automated loading
- Unified monitoring (PLG Stack: Prometheus · Loki · Grafana) and certificate automation (Cert-Manager + Let's Encrypt)
- Designed & built GPU infrastructure for image-generation AI workloads — 10 in-house boxes, GCP A100 and L4, 20 external-IDC servers

Key Achievements

- 50% Cloud Cost Reduction (Fargate → GKE node pools + MSP discount)
- 2-3h Daily Manual Collection Replaced (analyst hand work → automated loading)

Iorchard

2022.04 ~ 2023.03

Infra Engineer

Designed and built PaaS (Kubernetes + OpenStack + Ceph) private clouds for finance/public-sector clients; led operations training.

- Customized PaaS solution design and construction — delivered to 5 clients (Kubernetes HA + OpenStack + Ceph)
- DP (Delivery Partner) Kubernetes operations training program design and delivery (6 months)
- Wrote the planned shutdown and restart runbook for client power work, handed it to field engineers, and recovered restart failures as second-line support
- Operated and optimized OpenStack and Ceph storage clusters, with regular cluster inspection and troubleshooting
- Automated server provisioning with Ansible, building every client's servers from the same playbooks

Key Achievements

- 5 Cases Client PaaS Solution Deployments (Stable HA-based operation)
- 6 Months DP Field Engineer Kubernetes Training (Monthly · 4 hours a session · about 10 people)

ETech System

2022.01 ~ 2022.04

System Engineer

- Hardware Server and SAN storage construction
- Transitioned to DevOps field for career growth

Opensource Projects

Upstream Contributions (140+ merged, 80+ repos)

nginx/nginx-gateway-fabric · kubernetes-sigs/kueue · elastic/elasticsearch · apache/airflow

Kubernetes CRD

k8s-namespace-sync · helios-lb · network-policy-generator · kube-drift

Helm Charts

nginx-gateway-cr · karpenter-cr · aws-storageclass · buildkit

GitHub Actions

Image Tag Updater · Helm Chart Release Action · Go Kubebuilder Test Action · Helm OCI Push · Major Tag Action

Ansible Galaxy

Ansible K8s IAC Tool · Ansible User Management

DevOps Tools

git-bridge · static-file-server · kube-diff · kube-ctx · helm-upgrade-sync

Claude Code Plugins

sensitive-guard · release-guards · shell-portability

Education

Chungnam National University - Public Administration (Transfer)	2018.03 ~ 2020.08
National Institute for Lifelong Education - Business Administration (Academic Credit Bank)	2016.09 ~ 2018.02

Certifications

Engineer Information Processing	2021.11
AWS Certified Solution Architect - Associate (Expired)	2021.11
Network Administrator Level 2	2021.10
Linux Master Level 2	2021.10
Computer Specialist Level 1	2021.04
TOEIC Speaking Test - 130/Intermediate Mid 3 (Expired)	2021.02

Study

Tech blog (somaz.blog, English) — DevOps · Kubernetes · IaC writing for global readers	2025.01 ~ 현재
Tech blog (somaz.tistory.com, Korean) — ongoing DevOps · Kubernetes · IaC learning and operations notes	2023.04 ~ 현재
T101(Terraform) Study	2023.08 ~ 2023.10
AEWS(EKS) Study	2023.04 ~ 2023.06
PKOS(Kubernetes) Study	2023.01 ~ 2023.02
Cloud Security Training	2022.02 ~ 2022.04
Security Solution Operation Specialist Training Program	2021.07 ~ 2021.12